NU Map

Generated April 4, 2026

PROGRAM

Major **Industrial Engineering and Computer Science, BSIE (Boston)**

CREDITS

49 SH completed + **116** SH planned / **140** SH required **35%**

NUPATH (12/12 fulfilled)

- ✓ **ND** Natural/Designed World
- ✓ **EI** Creative Express/Innov
- ✓ **IC** Interpreting Culture
- ✓ **FQ** Formal/Quant Reasoning
- ✓ **SI** Societies/Institutions
- ✓ **AD** Analyzing/Using Data
- ✓ **DD** Difference/Diversity
- ✓ **ER** Ethical Reasoning
- ✓ **WD** Adv Writing Disc
- ✓ **WI** Writing Intensive
- ✓ **EX** Integration Experience
- ✓ **CE** Capstone Experience

MAJOR REQUIREMENTS — Industrial Engineering and Computer Science, BSIE (Boston)

Required Engineering 6/6

- **All of (2/2)**
  - IE 2310 — Introduction to Industrial Engineering
  - IE 2311 — Recitation for IE 2310
- IE 4510
- IE 4516
- **All of (2/2)**
  - IE 4522 — Human-Machine Systems
  - IE 4523 — Lab for IE 4522
- IE 4525
- **All of (2/2)**
  - IE 4530 — Manufacturing Systems and Techniques
  - IE 4531 — Lab for IE 4530

Senior Capstone Design Project 2/2

- MEIE 4701
- MEIE 4702

Supplemental Credit 6/6

- IE 3412
- IE 4512
- IE 4515
- IE 4520
- GE 1501
- GE 1502

Computer Science Fundamental Courses 3/3

- [x] **All of (2/2)**
  - [x] CS 1800 — Discrete Structures
  - [x] CS 1802 — Seminar for CS 1800
- [ ] **All of (2/2)**
  - [ ] CS 2000 — Introduction to Program Design and Implementation
  - [ ] CS 2001 — Lab for CS 2000
- [x] **All of (2/2)**
  - [x] CS 2100 — Program Design and Implementation 1
  - [x] CS 2101 — Lab for CS 2100

Computer Science Required Courses 5/5

- [ ] CS 3000
- [ ] **All of (2/2)**
  - [ ] CS 3100 — Program Design and Implementation 2
  - [ ] CS 3101 — Lab for CS 3100
- [ ] CS 3650
- [ ] **One of (1/2)**
  - [ ] CS 4530
  - [ ] CS 4535
- [ ] **One of (1/2)**
  - [ ] CS 4700
  - [ ] CS 4730

Khoury Approved Electives 1/1

- [x] **8/6 SH from elective pool**
  - [x] CS 4180, CS 5330 (CS range)
  - [ ] Any CY 2000–9999
  - [ ] Any DS 2500–9999

Required Mathematics/Science 6/6

- [ ] **All of (2/2)**
  - [ ] CHEM 1151 — General Chemistry for Engineers
  - [ ] CHEM 1153 — Recitation for CHEM 1151
- [x] MATH 1341
- [x] MATH 1342
- [x] MATH 2321
- [ ] MATH 2341
- [ ] **All of (3/3)**
  - [ ] PHYS 1151 — Physics for Engineering 1
  - [ ] PHYS 1152 — Lab for PHYS 1151
  - [ ] PHYS 1153 — Interactive Learning Seminar for PHYS 1151

General Electives 0/0

- [x] PHYS 1161
- [x] PHYS 1162
- [x] PHYS 1163
- [x] PHYS 1165
- [x] PHYS 1166
- [x] PHYS 1167
- [x] CS 1990
- [x] ENGW 1111
- [x] ARCH 1310
- [x] POLS 1150
- [x] CS 1200
- [ ] MATH 3081
- [ ] MATH 3527
- [ ] ENGW 3302
- [ ] MATH 2331
- [ ] GE 1111
- [ ] GE 1110
- [ ] PHYS 2371
- [ ] PHYS 2372
- [ ] ARTF 1230

PLACED OUT (satisfies prerequisites, no credit)

CS 2000 CS 2001

SUBSTITUTIONS (course A placed, satisfies course B — credits count once)

MATH 3081 → satisfies **IE 3412**
GE 1111 → satisfies **GE 1502**
GE 1110 → satisfies **GE 1501**

NU Map: Course Schedule

Generated April 4, 2026

Incoming Credit 30 SH · completed

PHYS	PHYS 1161	Physics 1	ND	4 SH
PHYS	PHYS 1162	Lab for PHYS 1161	AD	1 SH
PHYS	PHYS 1163	Recitation for PHYS 1161		0 SH
PHYS	PHYS 1165	Physics 2	ND	4 SH
PHYS	PHYS 1166	Lab for PHYS 1165	AD	1 SH
PHYS	PHYS 1167	Recitation for PHYS 1165		0 SH
MATH	MATH 1341	Calculus 1 for Science and Engineering	FQ	4 SH
MATH	MATH 1342	Calculus 2 for Science and Engineering	FQ	4 SH
CS	CS 1990	Elective		4–4 SH
ENGW	ENGW 1111	First-Year Writing		4 SH
ARCH	ARCH 1310	Buildings and Cities, A Global History	DD IC	4 SH

Fall 2025 19 SH · completed

CS	CS 1800	Discrete Structures	FQ	4 SH
CS	CS 1802	Seminar for CS 1800		1 SH
CS	CS 2100	Program Design and Implementation 1	AD ND	4 SH
CS	CS 2101	Lab for CS 2100		1 SH
MATH	MATH 2321	Calculus 3 for Science and Engineering	FQ	4 SH
POLS	POLS 1150	American Government	SI	4 SH
CS	CS 1200	First Year Seminar		1 SH

Spring 2026 17 SH · in progress

CS	CS 3100	Program Design and Implementation 2	AD ND	4 SH
CS	CS 3101	Lab for CS 3100		1 SH
MATH	MATH 2341	Differential Equations and Linear Algebra for Engineering		4 SH
MATH	MATH 3081	Probability and Statistics	AD	4 SH
MATH	MATH 3527	Number Theory 1		4 SH

Summer A 2026 4 SH

ENGW	ENGW 3302	Advanced Writing in the Technical Professions	WD	4 SH

Summer B 2026 9 SH

MATH	**MATH 2331** Linear Algebra		4 SH
PHYS	**PHYS 1151** Physics for Engineering 1	ND	3 SH
PHYS	**PHYS 1152** Lab for PHYS 1151	AD	1 SH
PHYS	**PHYS 1153** Interactive Learning Seminar for PHYS 1151		1 SH

Fall 2026 16 SH

CS	**CS 3000** Algorithms and Data	FQ	4 SH
CS	**CS 3001** Recitation for CS 3000		0 SH
IE	**IE 2310** Introduction to Industrial Engineering	WI	4 SH
IE	**IE 2311** Recitation for IE 2310		0 SH
IE	**IE 4515** Operations Research		4 SH
GE	**GE 1111** Engineering Problem Solving and Computation	FQ	4 SH

Spring 2027 17 SH

CS	**CS 3650** Computer Systems	4 SH
IE	**IE 4510** Simulation Modeling and Analysis	4 SH
IE	**IE 4525** Logistics and Supply Chain Management	4 SH
IE	**IE 4530** Manufacturing Systems and Techniques	4 SH
IE	**IE 4531** Lab for IE 4530	1 SH

Summer A 2027

Full-Time Internship · Google
Software Engineer
Spans into Summer B 2027 · 4-month block

Summer B 2027

Internship Continues · Google
Software Engineer
4-month block

Fall 2027 16 SH

CS	**CS 4180** Reinforcement Learning			4 SH
CS	**CS 4530** Fundamentals of Software Engineering	CE	WI	4 SH
IE	**IE 4516** Quality Assurance			4 SH
PHYS	**PHYS 2371** Electronics	ND		3 SH
PHYS	**PHYS 2372** Lab for PHYS 2371	EI		1 SH

Spring 2028

CO-OP · Amazon
Industrial Engineer
Spans into Summer A 2028 · 6-month block

Summer A 2028

CO-OP CONTINUES · Amazon
Industrial Engineer
6-month block

Fall 2028 17 SH

CS	**CS 4730**	Distributed Systems		4 SH
CS	**CS 5330**	Pattern Recognition and Computer Vision		4 SH
GE	**GE 1110**	Engineering Design	ER	4 SH
IE	**IE 4512**	Engineering Economy		4 SH
MEIE	**MEIE 4701**	Capstone Design 1	CE EI WI	1 SH

Spring 2029 20 SH

IE	**IE 4522**	Human-Machine Systems		4 SH
IE	**IE 4523**	Lab for IE 4522		1 SH
CHEM	**CHEM 1151**	General Chemistry for Engineers	ND	4 SH
CHEM	**CHEM 1153**	Recitation for CHEM 1151		0 SH
IE	**IE 4520**	Stochastic Modeling		4 SH
MEIE	**MEIE 4702**	Capstone Design 2	CE EI WI	5 SH
ARTF	**ARTF 1230**	Making with Form and Materials	EI	2 SH

Course Descriptions

PHYS **PHYS 1161** Physics 1 4 SH

Covers calculus-based physics. Offers the first semester of a two-semester integrated lecture and laboratory sequence intended primarily for science students. Covers Newtonian mechanics and fluids. Emphasizes the underlying concepts and principles. Takes applications from a wide variety of fields, such as life sciences and medicine, astro- and planetary physics, and so on. Includes topics such as forces, torque and static equilibrium, one-dimensional and three-dimensional motion, Newton's laws, dynamics friction, drag, work, energy and power, momentum and collisions, rotational dynamics, oscillations, pressure, fluids, and gravity.

PHYS **PHYS 1162** Lab for PHYS 1161 1 SH

Accompanies PHYS 1161. Covers topics from the course through various experiments.

PHYS **PHYS 1163** Recitation for PHYS 1161 0 SH

Accompanies PHYS 1161. Offers an opportunity for interactive problem solving.

PHYS **PHYS 1165** Physics 2 4 SH

Continues PHYS 1161. Offers the second semester of a two-semester integrated lecture and laboratory sequence intended primarily for science students. Includes topics such as electrostatics; capacitors; resistors and direct-current circuits; magnetism and magnetic induction; RC, LR, and LRC circuits; waves; electromagnetic waves; and fluids.

PHYS **PHYS 1166** Lab for PHYS 1165 1 SH

Accompanies PHYS 1165. Covers topics from the course through various experiments.

PHYS **PHYS 1167** Recitation for PHYS 1165 0 SH

Accompanies PHYS 1165. Offers an opportunity for interactive problem solving.

MATH **MATH 1341** Calculus 1 for Science and Engineering 4 SH

Covers definition, calculation, and major uses of the derivative, as well as an introduction to integration. Topics include limits; the derivative as a limit; rules for differentiation; and formulas for the derivatives of algebraic, trigonometric, and exponential/logarithmic functions. Also discusses applications of derivatives to motion, density, optimization, linear approximations, and related rates. Topics on integration include the definition of the integral as a limit of sums, antidifferentiation, the fundamental theorem of calculus, and integration by substitution.

MATH **MATH 1342** Calculus 2 for Science and Engineering 4 SH

Covers further techniques and applications of integration, infinite series, and introduction to vectors. Topics include integration by parts; numerical integration; improper integrals; separable differential equations; and areas, volumes, and work as integrals. Also discusses convergence of sequences and series of numbers, power series representations and approximations, 3D coordinates, parameterizations, vectors and dot products, tangent and normal vectors, velocity, and acceleration in space.

CS **CS 1990** Elective 4 SH

Offers elective credit for courses taken at other academic institutions. May be repeated without limit.

ENGW **ENGW 1111** First-Year Writing 4 SH

Designed for students to study and practice writing in a workshop setting. Students read a range of texts in order to describe and evaluate the choices writers make and apply that knowledge to their own writing and explore how writing functions in a range of academic, professional, and public contexts. Offers students an opportunity to learn how to conduct research using primary and secondary sources; how to write for various purposes and audiences in multiple genres and media; and how to give and receive feedback, to revise their work, and to reflect on their growth as writers.

ARCH **ARCH 1310** Buildings and Cities, A Global History 4 SH

Introduces students to architecture, as understood through buildings, cities, and landscapes from antiquity to the present. Studies important monuments in the global history of architecture, as well as tools for analyzing the built environment. Considers buildings in relation to their political, social, economic, and cultural context, and as expressions of diversity in human societies and cultural perspectives. Topics include the language of architecture, architectural drawings, the classical orders, the problem of ornament, construction techniques, materials, site,and the role of the patron. Develops students' eye for composition in two and three dimensions, aesthetic discrimination of detail, ability to see buildings as part of a larger social and cultural fabric, and critical judgment in speaking and writing.

CS **CS 1800** Discrete Structures 4 SH

Introduces the mathematical structures and methods that form the foundation of computer science. Studies structures such as sets, tuples, sequences, lists, trees, and graphs. Discusses functions, relations, ordering, and equivalence relations. Examines inductive and recursive definitions of structures and functions. Discusses principles of proof such as truth tables, inductive proof, and basic logic. Also covers the counting techniques and arguments needed to estimate the size of sets, the growth of functions, and the space-time complexity of algorithms.

CS **CS 1802** Seminar for CS 1800 1 SH

Accompanies CS 1800. Illustrates topics from the lecture course through discussions, quizzes, and homework assignments.

CS **CS 2100** Program Design and Implementation 1 4 SH

Builds on prior introductory programming experience (from inside or outside Northeastern University) to examine the fundamentals of program design and implementation. Studies design of data- and object-oriented programs including common patterns; use of data structures; and underlying principles such as abstraction, encapsulation, inheritance, and interfaces. Introduces common software engineering practices such as test-driven development, version control, development environments, and good programming habits. Students practice using these design principles by writing medium-sized applications and using data science libraries. Covers skills needed to identify and respond to ethical challenges that arise in the program design process.

CS **CS 2101** Lab for CS 2100 1 SH

Accompanies CS 2100. Covers topics from the course through various experiments.

MATH **MATH 2321** Calculus 3 for Science and Engineering 4 SH

Extends the techniques of calculus to functions of several variables; introduces vector fields and vector calculus in two and three dimensions. Topics include lines and planes, 3D graphing, partial derivatives, the gradient, tangent planes and local linearization, optimization, multiple integrals, line and surface integrals, the divergence theorem, and theorems of Green and Stokes with applications to science and engineering and several computer lab projects. Requires prior completion of MATH 1342 or MATH 1252.

POLS **POLS 1150** American Government 4 SH

Analyzes the system of politics and government in the United States. Topics include the philosophical basis, historical origins, design, and functioning of the Constitution as well as formal government institutions. Examines the influence of public opinion, political behavior and participation, parties, and interest groups.

CS **CS 1200** First Year Seminar 1 SH

Seeks to support students in their transition to Northeastern and in their holistic development as they become responsible members of the college and university communities. Incorporates large group discussion, small group activities, and self-reflection in order to facilitate connections with faculty, staff, and peers; promote utilization of appropriate campus resources; and assist with academic and personal goal setting.

CS **CS 3100** Program Design and Implementation 2 4 SH

Builds on foundations introduced in CS 2100 to examine program design at increasing scales of complexity. Reviews abstraction, encapsulation, inheritance, and interfaces in statically typed object-oriented languages. Presents a comparative approach to software design patterns and paradigms, including object-oriented and functional programming. Offers students an opportunity to obtain a deeper understanding of the principles of program design including interface design, test-driven development, graphical design notations, reusable software components, and open-source ecosystems. Illustrates the impact of design-time decisions on software correctness including accessibility, changeability, performance, reusability, and privacy. Students collaborate throughout the semester to design and implement a large software project.

CS **CS 3101** Lab for CS 3100 1 SH

Accompanies CS 3100. Covers topics from the course and provides students with additional opportunities to engage with course material.

MATH **MATH 2341** Differential Equations and Linear Algebra for Engineering 4 SH

Studies ordinary differential equations, their applications, and techniques for solving them including numerical methods (through computer labs using MS Excel and MATLAB), Laplace transforms, and linear algebra. Topics include linear and nonlinear first- and second-order equations and applications include electrical and mechanical systems, forced oscillation, and resonance. Topics from linear algebra, such as matrices, row-reduction, vector spaces, and eigenvalues/eigenvectors, are developed and applied to systems of differential equations. Requires prior completion of MATH 1342.

MATH **MATH 3081** Probability and Statistics 4 SH

Focuses on probability theory. Topics include sample space; conditional probability and independence; discrete and continuous probability distributions for one and for several random variables; expectation; variance; special distributions including binomial, Poisson, and normal distributions; law of large numbers; and central limit theorem. Also introduces basic statistical theory including estimation of parameters, confidence intervals, and hypothesis testing.

MATH **MATH 3527** Number Theory 1 4 SH

Introduces number theory. Topics include linear diophantine equations, congruences, design of magic squares, Fermat's little theorem, Euler's formula, Euler's phi function, computing powers and roots in modular arithmetic, the RSA encryption system, primitive roots and indices, and the law of quadratic reciprocity. As time permits, may cover diophantine approximation and Pell's equation, elliptic curves, points on elliptic curves, and Fermat's last theorem.

ENGW **ENGW 3302** Advanced Writing in the Technical Professions 4 SH

Offers writing instruction for students in the College of Engineering and the College of Computer and Information Science. Students practice and reflect on writing in professional, public, and academic genres—such as technical reports, progress reports, proposals, instructions, presentations, and technical reviews—relevant to technical professions and individual student goals. In a workshop setting, offers students an opportunity to evaluate a wide variety of sources and develop expertise in audience analysis, critical research, peer review, and revision.

MATH **MATH 2331** Linear Algebra 4 SH

Uses the Gauss-Jordan elimination algorithm to analyze and find bases for subspaces such as the image and kernel of a linear transformation. Covers the geometry of linear transformations: orthogonality, the Gram-Schmidt process, rotation matrices, and least squares fit. Examines diagonalization and similarity, and the spectral theorem and the singular value decomposition. Is primarily for math and science majors; applications are drawn from many technical fields. Computation is aided by the use of software such as Maple or MATLAB, and graphing calculators.

PHYS **PHYS 1151** Physics for Engineering 1 3 SH

Students registered for PHYS 1151 must also be registered for PHYS 1152 and PHYS 1153.

PHYS **PHYS 1152** Lab for PHYS 1151 1 SH

Accompanies PHYS 1151. Covers topics from the course through various experiments. Requires concurrent registration in PHYS 1151 and PHYS 1153.

PHYS **PHYS 1153** Interactive Learning Seminar for PHYS 1151 1 SH

Students registered for PHYS 1153 must also be registered for PHYS 1151 and PHYS 1152.

CS **CS 3000** Algorithms and Data 4 SH

Introduces the basic principles and techniques for the design, analysis, and implementation of efficient algorithms and data representations. Discusses asymptotic analysis and formal methods for establishing the correctness of algorithms. Considers divide-and-conquer algorithms, graph traversal algorithms, and optimization techniques. Introduces information theory and covers the fundamental structures for representing data. Examines flat and hierarchical representations, dynamic data representations, and data compression. Concludes with a discussion of the relationship of the topics in this course to complexity theory and the notion of the hardness of problems.

CS **CS 3001** Recitation for CS 3000 0 SH

Accompanies CS 3000. Provides students with additional opportunities to ask questions and engage with course material.

IE **IE 2310** Introduction to Industrial Engineering 4 SH

Provides an overview of the history of industrial engineering and of the most common methods that industrial engineers use to solve problems and design efficient processes. The emphasis is on how these methods are used to study, improve, and/or optimize a product or process. Topics include work design, ergonomic design, engineering statistics, quality engineering, engineering economics, project management, and process optimization. Also discusses the design of the production processes, facilities, and material handling systems. Studies applications in manufacturing, product design, and service industries. Laboratory experiments and written reports are required.

IE **IE 2311** Recitation for IE 2310 0 SH

Provides small group demonstration and hands-on labs for IE 2310.

IE **IE 4515** Operations Research 4 SH

Introduces deterministic models including linear programming; duality and postoptimality analysis; transportation and assignment problems; and network flow problems such as the shortest path, minimum spanning tree, and maximum flow.

GE **GE 1111** Engineering Problem Solving and Computation 4 SH

Uses a structured approach to solve engineering problems. Draws applications from a variety of engineering disciplines, which serve as a tool for introducing students to engineering analysis and design. Introduces a math application package for matrix applications and various real-life engineering problems. Includes the design of problem-solving algorithms using a high-level programming language. Topics include the use of programmable microcontrollers, as well as various electronic components.

CS **CS 3650** Computer Systems 4 SH

Introduces the basic design of computing systems, computer operating systems, and assembly language using a RISC architecture. Describes caches and virtual memory. Covers the interface between assembly language and high-level languages, including call frames and pointers. Covers the use of system calls and systems programming to show the interaction with the operating system. Covers the basic structures of an operating system, including application interfaces, processes, threads, synchronization, interprocess communication, deadlock, memory management, file systems, and input/output control.

IE **IE 4510** Simulation Modeling and Analysis 4 SH

Covers process model design and development, validation, and experimentation for discrete-event simulation models. Topics include problem formulation, data collection and analysis, random-variable generation, model development, scenario experimentation, statistical analysis of output, and resultant decision management. Utilizes a major industry-standard simulation software application with animation capabilities.

IE **IE 4525** Logistics and Supply Chain Management 4 SH

Introduces the analysis, design, control, and operation of logistics and supply chain management systems. Includes the integration of supply chain components, logistics information systems, forecasting, production scheduling, inventory management, transportation and warehousing, and facility location planning.

IE **IE 4530** Manufacturing Systems and Techniques 4 SH

Focuses on manufacturing and design and their impact on each other. Covers the basics of design-manufacturing integration, manufacturing systems, manufacturing processes and techniques, manufacturing automation, and production planning and control. Topics include concurrent engineering, design for assembly, design for manufacturability, rapid prototyping, mechanical tolerancing, bill of materials, group technology, computer-aided process planning, NC part programming, programmable logic controllers, flexible manufacturing systems, computer-integrated manufacturing, and just-in-time philosophy. Topics also include traditional manufacturing processes such as casting, forming, machining, welding, molding, and particulate processing, and nontraditional manufacturing processes such as electrical discharge machining, laser machining, and water-jet machining. Students are required to conduct manufacturing-related experiments in the manufacturing lab to gain hands-on experience.

IE **IE 4531** Lab for IE 4530 1 SH

Accompanies IE 4530. Covers topics from the course through various activities.

CS **CS 4180** Reinforcement Learning 4 SH

Introduces reinforcement learning and the Markov decision process (MDP) framework. Covers methods for planning and learning in MDPs such as dynamic programming, model-based methods, and model-free methods. Examines commonly used representations including deep-learning representations. Students are expected to have a working knowledge of probability, to complete programming assignments, and to complete a course project that applies some form of reinforcement learning to a problem of interest.

CS **CS 4530** Fundamentals of Software Engineering 4 SH

Covers the fundamentals of software engineering, including software development life cycle models (e.g., waterfall, spiral,agile); requirements analysis; user-centered design; software design principles and patterns; testing (functional testing, structural testing, testing strategies); code refactoring and debugging; software architecture and design; and integration and deployment. Includes a course project in which some of the software engineering methods (from requirements analysis to testing) are applied in a team-based setting.

IE **IE 4516** Quality Assurance 4 SH

Reviews the distributions and statistical approximations commonly applied in statistical quality control methods. Introduces analysis of variance and simple linear regression. Covers basic principles to state-of-the-art concepts and application of statistical process control and design. Applies principles to a variety of products. Topics include product quality measures and controls, Shewhart control charts, quality cost, Pareto analysis, discrete and variable sampling, and military standards in quality control.

IE **IE 4522** Human-Machine Systems 4 SH

Emphasizes and addresses human sensory and motor performance, information processing, learning and training methodology, skilled-task development, psychophysical models, response time, and relevant aspects of attention and memory. Topics include system design and development, hazard and error evaluation, and properties of effective visual displays. Endorses experimentation as a source of knowledge of human performance characteristics. Covers research and statistical analyses related to human-asset engineering, fundamentals of vision, audition, somesthesis, signal detection, and some aging effects. Safety and usability of environments, machines, products, and devices consider principles of human-machine interaction, decision making, and anthropometric characteristics. Laboratory experiences include literature review, experimental design, data collection and analysis, hypothesis testing, and generation of reports to inform the design of safe, usable, and marketable engineering products, processes, and systems.

IE **IE 4523** Lab for IE 4522 1 SH

Accompanies IE 4522. Covers topics from the course through various activities.

CS **CS 4730** Distributed Systems 4 SH

Introduces distributed systems, covering fundamental concepts and showing how they are applied to build reliable distributed services. Examines several existing distributed applications, such as file systems, databases, lock services, digital currencies, smart contracts, and machine learning, and how these applications must coordinate to function and overcome failures, network partitions, or compromised parties. Distributed systems, such as databases, cloud services, and blockchains, are omnipresent in the services and applications that serve society on a daily basis.

CS **CS 5330** Pattern Recognition and Computer Vision 4 SH

Section meets 2-5:20pm PT/ 5-8:20pm ET

GE **GE 1110** Engineering Design 4 SH

Seeks to develop problem-solving skills used in engineering design, using case studies for a variety of engineering disciplines. Introduces students to the use of spreadsheet tools to solve engineering problems, including data reduction and visualization of data and functions. Design topics include problem formulation and specification, creativity, evaluation tools, patents, ergonomics, system design, manufacturing, ethics in engineering, and presentation techniques. Presents engineering graphics focusing on developing 3D visualization skills and computer-aided design (CAD) application. Students develop an original design solution to a technical problem as a term project. Requires students to have a laptop computer that meets the specifications of the College of Engineering.

IE **IE 4512** Engineering Economy 4 SH

Introduces students to economic modeling and analysis techniques for selecting alternatives from potential solutions to an engineering problem. Presents basic methods of economic comparison such as present worth, annual worth, rate of return, and benefit/cost techniques. Studies effects of taxes on investment analysis. Also covers decision tree analysis and statistical decision techniques.

MEIE **MEIE 4701** Capstone Design 1 1 SH

Offers the first in a two-course sequence that culminates the student's education and experience with the design process. Students form teams and are assigned their design project and faculty adviser. Projects can be industrially, departmentally, or externally sponsored. Students are expected to communicate with their faculty adviser, course coordinator, and sponsor using the Internet, teleconferencing, and other electronic methods. Topics include project management, ethics, cost analysis, Internet and library research methods, and engineering codes and standards. Students prepare written reports and make oral presentations. Students are expected to complete a thorough state-of-the-art report on their problem and a problem statement with specifications and requirements.

CHEM **CHEM 1151** General Chemistry for Engineers 4 SH

Students registering for this lecture must register for one of the following sections of the recitation CHEM 1153: 10215 (section 5), 10233 (section 14), and 10232 (section 15)

CHEM **CHEM 1153** Recitation for CHEM 1151 0 SH

Students registering for this recitation must register for one of the following lecture sections of CHEM 1151: 13213 (section 5)

IE **IE 4520** Stochastic Modeling 4 SH

Covers the analytical development and solution to stochastic models in operations research. Topics include Markov chains, queuing theory, and dynamic programming.

MEIE **MEIE 4702** Capstone Design 2 5 SH

Continues MEIE 4701. Students are expected to apply engineering principles acquired throughout their undergraduate academic and co-op experiences to the design of a system, component, or process. Each project includes the development and use of design methodology, formulation of design problem statements and specifications, consideration of alternative solutions, feasibility considerations, and detailed system descriptions. Projects include realistic constraints such as economic factors, safety, reliability, maintenance, aesthetics, ethics, and political and social impact. Students make oral presentations on their results in a series of design reviews. Students document their solutions using a written report that includes an executive summary. A working prototype or simulation, as appropriate, of their solution is required to complete the course.

PHYS **PHYS 2371** Electronics 3 SH

Covers the physics underlying computers and our modern electronic world. Focuses on principles of semiconductor devices (diodes, transistors, integrated circuits, LEDs, photovoltaics); analog techniques (amplification, AC circuits, resonance); digital techniques (binary numbers, NANDs, logic gates, and circuits); electronic subsystems (operational amplifiers, magnetoelectronics, optoelectronics); and understanding commercial electronic equipment. Lab experiments are designed to investigate the properties of discrete and integrated devices and use them to design and build circuits.

PHYS **PHYS 2372** Lab for PHYS 2371 1 SH

Accompanies PHYS 2371. Illustrates topics from the lecture course through various hands-on experimental projects. Covers the process of electronics design from a goal-oriented perspective. Students are expected to consider their own electronics design project and build a prototype device that accomplishes a specific purpose.

ARTF **ARTF 1230** Making with Form and Materials 2 SH

Introduces concepts, processes, and studio practice in 3D making. Examines the elements and principles of 3D making, including mass, volume, line, plane, and texture. Hands-on experimentation offers students an opportunity to develop skills to define 3D form, express ideas through it, and solve visual problems. Students experiment with additive, subtractive, and substitutive material processes while leveraging both traditional craft techniques and digital fabrication. Focuses on developing a practical understanding of the relationship between material, form, function, aesthetic and context, and a vocabulary to successfully articulate concepts and insights during discussions and critiques.